

HAM10000: Five-Notebook Comparison

Top-to-bottom audit of the uploaded ResNet50, EfficientNet-B2, ViT-B/16, Swin-Tiny and GNN notebooks. Comparison covers data protocol, leakage control, augmentation, model/training setup, imbalance handling, validation strategy, final metrics, class-wise behavior, explainability, ablation, uncertainty and reporting caveats.

Fair-comparison rule

ViT-B/16, Swin-Tiny and the final GNN use the same lesion-level final test split (seed 171, 1,506 images). ResNet50 and EfficientNet-B2 use a different seed-42 lesion-level holdout (1,494 test images). Therefore, raw metrics for all five are shown, but only the three seed-171 results are directly same-test comparable.

Executive performance snapshot

Notebook / final model	Test protocol	Test n	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Macro ROC-AUC	Directly same-test?
ResNet50	Seed 42; lesion holdout	1,494	0.7851	0.5672	0.6918	0.5887	0.7805	Not available*	No
EfficientNet-B2	Seed 42; lesion holdout	1,494	0.8052	0.5956	0.6638	0.6152	0.8050	Not available*	No
ViT-B/16 (5-fold ensemble)	Seed 171; fixed final test	1,506	0.8652	0.7772	0.7859	0.7777	0.8677	0.9772**	Yes
Swin-Tiny (5-fold ensemble)	Seed 171; fixed final test	1,506	0.8546	0.7632	0.7395	0.7289	0.8581	0.9729	Yes
GNN - final GATv2	Seed 171; fixed final test	1,506	0.7543	0.5347	0.5668	0.5453	0.7621	0.8806	Yes

* ResNet50 and EfficientNet-B2 notebooks caught a ROC-AUC ValueError during final evaluation and stored the aggregate AUC as None. ** The ViT main metric summary stored macro AUC as NaN, but the later class-wise one-vs-rest ROC section reports seven valid AUCs; 0.9772 is their arithmetic mean, clearly marked as derived rather than copied from the main summary.

Same-test winner

ViT-B/16 ensemble: highest accuracy (86.52%) and Macro F1 (0.7777) among ViT/Swin/GNN.

Most parameter-efficient CNN

EfficientNet-B2: 7.71M parameters and better raw seed-42 test results than ResNet50.

Most extensive robustness workflow

ViT/Swin: lesion-safe 5-fold CV plus fixed external-to-fold test ensemble. GNN adds ablation + lesion bootstrap CI.

1. Notebook scan coverage and data protocol

Notebook	Cells	Code / Markdown	Stored outputs	Stored error outputs	Main workflow
ResNet50	32	17 / 15	174	0	Single lesion-level train/val/test split; head warm-up then full fine-tuning; best checkpoint by validation Macro F1.
EfficientNet-B2	33	17 / 16	175	0	Same overall pipeline as ResNet50, with EfficientNet-B2 input size/model head.
ViT-B/16	69	35 / 34	113	0	Fixed final test + lesion-safe 5-fold CV; fold checkpoints; probability ensemble; attention explainability.
Swin-Tiny	60	31 / 29	146	0	Fixed final test + lesion-safe 5-fold CV; fold checkpoints; ensemble; Grad-CAM/error analysis.
GNN	60	33 / 27	90	0	Fixed ResNet50 spatial feature encoder; image-to-grid graph; sequential GNN ablation; final GATv2 test; bootstrap + SHAP/LIME.

Aspect	ResNet50	EfficientNet-B2	ViT-B/16	Swin-Tiny	GNN
Dataset	HAM10000; 10,015 images; 7 classes	Same	Same	Same	Same
Leakage unit	lesion_id	lesion_id	lesion_id	lesion_id	lesion_id
Primary split	GroupShuffleSplit; ~70/15/15	Same	15% fixed test; remaining dev for CV	Same as ViT	15% fixed test, then train/val from development
Split seed / selection	Seed 42	Seed 42	200 candidate group splits; selected seed 171	Same seed 171 test	Same seed 171 test; val candidate search selected seed 122
Train / Val / Test images	7,002 / 1,519 / 1,494	7,002 / 1,519 / 1,494	Dev 8,509 / Test 1,506; 5 CV folds inside dev	Dev 8,509 / Test 1,506; same folds	7,007 / 1,502 / 1,506
Train / Val / Test lesions	5,229 / 1,120 / 1,121	5,229 / 1,120 / 1,121	Dev 6,349 / Test 1,121	Same	5,228 / 1,121 / 1,121
5-fold CV	No	No	Yes - StratifiedGroupKFold	Yes - StratifiedGroupKFold	No (explicitly)
Final ensemble	No	No	Mean probability of 5 fold models	Mean probability of 5 fold models	No; final selected GNN only
Leakage checks	No lesion overlap reported	No lesion overlap reported	No dev/test lesion overlap + fold group checks	Same	No lesion/image overlap reported

Class imbalance context: HAM10000 is strongly dominated by nv (6,705/10,015). All notebooks explicitly address imbalance, but with different mechanisms (weighted loss vs weighted sampler).

2. Architecture, preprocessing and training configuration

Aspect	ResNet50	EfficientNet-B2	ViT-B/16	Swin-Tiny	GNN
Backbone / variant	timm ResNet50; pretrained	timm EfficientNet-B2; pretrained	vit_base_patch16_224.augreg_in21k_ft_in1k	swin_tiny_patch4_window7_224.ms_in22k_ft_in1k	Fixed torchvision ResNet50 encoder + final GATv2 graph classifier
Input image size	224 x 224	260 x 260	224 x 224	224 x 224	224 x 224 for CNN feature extraction
Reported model params	23.522M	7.711M	85.804M	27.525M	GNN parameter total not explicitly reported; fixed CNN encoder used upstream
Classifier head	Linear 2048 -> 7	Linear 1408 -> 7	Linear 768 -> 7	Linear 768 -> 7	Graph layers -> global mean pool -> MLP classifier
CNN/transformer batch	16	16	16	16	CNN encoder batch 32
GNN batch	-	-	-	-	64
Epoch budget	50 total	50 total	Max 50 per fold	Max 50 per fold	Encoder max 50; GNN max 50
Head-only warm-up	5 epochs	5 epochs	5 epochs	5 epochs	Encoder warm-up 5 epochs
Full/partial unfreeze	All backbone after epoch 5	All backbone after epoch 5	All backbone after warm-up	All backbone after warm-up	Encoder: unfreeze layer4 + head; GNN trained on cached spatial features
AMP	Yes when CUDA	Yes when CUDA	Yes	Yes; scaler overflow warnings seen, training continued	Used in CNN training pipeline where configured

Preprocessing / augmentation	ResNet50	EfficientNet-B2	ViT-B/16	Swin-Tiny	GNN encoder
Train crop	RandomResizedCrop; scale .80-1.00	Same, 260	RandomResizedCrop; scale .75-1.00; ratio .90-1.10	RandomResizedCrop; scale .80-1.00; ratio .90-1.10	RandomResizedCrop style, 224
Flips / rotation	H .5, V .5, rotation 20	Same	H .5, V .5, rotation 20	Same	H/V + rotation used
Color jitter	.20/.20/.20/.05	Same	.15/.15/.12/.02	.20/.20/.20/.05	Color augmentation used
Extra train aug.	-	-	GaussianBlur p=.10	-	-
Eval preprocessing	Resize 224	Resize 260	Resize 256 + CenterCrop 224	Bicubic resize 224 x 224	Deterministic resize/normalize
Normalization	ImageNet mean/std	ImageNet mean/std	ImageNet mean/std	ImageNet mean/std	ImageNet-compatible encoder normalization

3. Optimization, imbalance handling and model selection

Aspect	ResNet50	EfficientNet-B2	ViT-B/16	Swin-Tiny	GNN
Imbalance strategy	Sqrt inverse class-weighted CE	Same	WeightedRandomSampler + CE	WeightedRandomSampler + CE	Sqrt-balanced weighted CE; ablation also considers focal/plain CE
Label smoothing	0.10	0.10	0.10 train; 0 val	0.10 train; 0 val	0.05 for encoder/GNN weighted CE
Optimizer	AdamW	AdamW	AdamW; decay/no-decay parameter groups	AdamW; decay/no-decay groups	Encoder AdamW; final GNN selected Adam
Warm-up/head LR	1e-3	1e-3	3e-4	3e-4	Encoder 3e-4
Fine-tune backbone LR	1e-5	1e-5	1e-5	1e-5	Encoder layer4 1e-5
Fine-tune head LR	1e-4	1e-4	5e-5	5e-5	Encoder head 1e-4; GNN final 2e-4
Weight decay	1e-4	1e-4	0.05	0.05	Encoder 1e-4; GNN config optimized separately
Scheduler	CosineAnnealingLR	CosineAnnealingLR	Linear warm-up -> cosine	Step-level linear warm-up -> cosine	Encoder ReduceLROnPlateau; final GNN cosine
Gradient clipping	Not reported	Not reported	Max norm 1.0	Max norm 1.0	Not central in final GNN config
Early stopping	No; all 50 epochs	No; all 50 epochs	Start 15; patience 8; min delta .001	Start 15; patience 8; min delta .001	Encoder patience 8; final GNN selected patience 5
Checkpoint selection	Best validation Macro F1	Best validation Macro F1	Best validation Macro F1 per fold	Best validation Macro F1 per fold	Ablation/selection by validation Macro F1

Fold / model	Best epoch	Completed epochs	Best Val Macro F1	Test Accuracy	Test Macro F1
ViT fold 1	25	33	0.7500	0.8533	0.7470
ViT fold 2	31	33	0.7806	0.8552	0.7598
ViT fold 3	17	25	0.7673	0.8400	0.7412
ViT fold 4	30	38	0.8148	0.8493	0.7522
ViT fold 5	13	22	0.7793	0.7875	0.7089
Swin fold 1	15	23	0.7155	0.8433	0.7152
Swin fold 2	25	33	0.7620	0.8533	0.7272
Swin fold 3	15	23	0.7009	0.8094	0.6911
Swin fold 4	33	35	0.7596	0.8307	0.6892
Swin fold 5	20	28	0.7588	0.8340	0.6909

ViT CV validation Macro F1 mean +/- std: 0.7784 +/- 0.0238. Swin pooled OOF Macro F1: 0.7404; fold-mean Macro F1: 0.7394 +/- 0.0289.

4. Final test metrics and class-wise behavior

Model	Accuracy	Balanced Acc.	Macro P	Macro R	Macro F1	Weighted F1	Macro AUC
ResNet50	0.7851	Not separately reported	0.5672	0.6918	0.5887	0.7805	N/A*
EfficientNet-B2	0.8052	Not separately reported	0.5956	0.6638	0.6152	0.8050	N/A*
ViT-B/16 ensemble	0.8652	Not separately reported	0.7772	0.7859	0.7777	0.8677	0.9772**
Swin-Tiny ensemble	0.8546	Not separately reported	0.7632	0.7395	0.7289	0.8581	0.9729
GNN final GATv2	0.7543	0.5668	0.5347	0.5668	0.5453	0.7621	0.8806

Class (support on seed-171 test)	ResNet50 F1*	EfficientNet-B2 F1*	ViT ensemble F1	Swin ensemble F1	GNN final F1
akiec (50)	0.5030	0.5441	0.6742	0.4857	0.3478
bcc (76)	0.5309	0.6571	0.8047	0.7927	0.5233
bkl (164)	0.5783	0.6199	0.7864	0.7714	0.5681
df (15)	0.4444	0.3200	0.6452	0.6061	0.3684
mel (167)	0.4784	0.5425	0.6829	0.6850	0.5366
nv (1,014)	0.9072	0.9146	0.9277	0.9281	0.8780
vasc (20)	0.6786	0.7083	0.9231	0.8333	0.5946

* ResNet50/EfficientNet-B2 class-wise F1 values come from their different 1,494-image seed-42 test set; the class support shown in the first column belongs to the shared 1,506-image seed-171 test and therefore must not be applied to those two columns. Within the genuinely shared test, ViT is strongest on akiec/bcc/bkl/df/vasc; Swin is marginally higher on mel and nv.

Class-wise OvR AUC (shared 1,506-image test)	ViT	Swin-Tiny	GNN
akiec	0.9714	0.9663	0.8098
bcc	0.9883	0.9913	0.9027
bkl	0.9728	0.9714	0.8860
df	0.9920	0.9641	0.8598
mel	0.9522	0.9505	0.8808
nv	0.9654	0.9708	0.9284
vasc	0.9983	0.9956	0.8963
Mean	0.9772	0.9729	0.8806

5. GNN-specific pipeline, ablation and uncertainty

Component	What the notebook does	Key values / result
Feature encoder	Pretrained ResNet50 image encoder; head warm-up then layer4 + head fine-tuning; features cached for GNN experiments.	Best encoder validation Macro F1 = 0.6295 at epoch 21; early stop at 29.
Graph construction	Extracts spatial feature map from layer4[-1].bn2; each image becomes a 7 x 7 grid of 49 nodes. Normalized row/column coordinates are concatenated to CNN features.	Node tensor shape: [N, 49, 514]. grid4 graph: 217 directed edges including self-loops; grid8: 361.
Candidate GNNs	Flexible implementation supports GCN, GATv2, GraphSAGE and SuperGAT; optional BatchNorm, residual graph layers, global mean pooling.	Final selected variant: GATv2, 2 layers, hidden 128, dropout 0.4, BatchNorm on, 4 heads.
Ablation design	Sequential one-factor-at-a-time stages rather than full Cartesian hyperparameter search. Each stage carries forward the selected setting.	Factors: layers, hidden size, dropout, BatchNorm, LR, scheduler, optimizer, graph method, edge weights, GNN variant, imbalance method, patience.
Initial -> selected validation	Initial GATv2 reference compared against sequentially improved configuration.	Initial val Macro F1 0.5443 / AUC 0.8906; best ablation record Macro F1 0.5928 / AUC 0.9020.
Final retrain/test	Final selected config retrained, then evaluated once on untouched seed-171 test.	Test accuracy 0.7543; Macro F1 0.5453; Macro AUC 0.8806. Training time 40.82 s; test 0.27 s.

Ablation stage / selected setting	Val Accuracy	Val Macro F1	Val Macro AUC
Reference initial config	0.7237	0.5443	0.8906
Layers = 2	0.7430	0.5798	0.8999
Scheduler = cosine	0.7450	0.5829	0.9014
Optimizer = Adam	0.7477	0.5854	0.9014
Patience = 5 (best ablation record)	0.7483	0.5928	0.9020

Final GNN metric	Point estimate	Lesion-bootstrap 95% CI
Accuracy	0.7543	[0.7293, 0.7790]
Balanced accuracy	0.5668	[0.5043, 0.6409]
Macro precision	0.5347	[0.4768, 0.5897]
Macro recall	0.5668	[0.5043, 0.6409]
Macro F1	0.5453	[0.4846, 0.5966]
Weighted F1	0.7621	[0.7374, 0.7858]
Macro ROC-AUC	0.8806	[0.8440, 0.9124]

6. Explainability, outputs and audit findings

Notebook	Explainability / analysis	Saved outputs / reporting	Notable audit finding
ResNet50	Training curves; confusion matrices; class report.	Best model, metrics, predictions, split files, plots; ZIP ~332.84 MB.	Aggregate ROC-AUC final calculation failed and was stored as None.
EfficientNet-B2	Training curves; confusion matrices; class report.	Best model, metrics, predictions, split files, plots; ZIP ~110.35 MB.	Aggregate ROC-AUC final calculation failed and was stored as None.
ViT-B/16	Attention rollout for all seven classes plus high-confidence misclassification examples.	Fold models/backup, predictions, OOF and test tables, CMs, ROC plots, result bundles.	Main Macro ROC-AUC summary is NaN, but later class-wise AUCs are valid; mean = 0.9772.
Swin-Tiny	Grad-CAM verification, class-wise examples, misclassification/error analysis.	5-fold artifacts, ensemble outputs, 106-file results bundle ~17.28 MB; compact backup ~487.97 MB.	AMP scaler overflow/optimizer-step warnings appear during training; no stored exception stopped execution.
GNN	SHAP + LIME node-importance explanations for one high-confidence correct and one incorrect test example.	Ablation tables, final metrics/predictions, bootstrap results, baseline comparison and explainability outputs.	Baseline significance fallback used unpaired aggregate 2-proportion z-test because per-image ViT prediction CSV was not found; notebook itself flags this as weaker than paired McNemar.

Audit conclusion	Evidence from notebooks	Interpretation
Best shared-test predictive result	ViT ensemble: accuracy 0.8652, Macro F1 0.7777; Swin: 0.8546 / 0.7289; GNN: 0.7543 / 0.5453.	On the identical seed-171 test, ViT is the strongest of the three directly comparable pipelines.
Best CNN on seed-42 holdout	EfficientNet-B2 > ResNet50 in raw accuracy (0.8052 vs 0.7851) and Macro F1 (0.6152 vs 0.5887).	This is a valid comparison between those two because they use the same 1,494-image split.
Minority-class sensitivity	df has only 15 examples on the shared test; vasc 20; akiec 50. Class F1 varies substantially by architecture.	Macro metrics are more informative than accuracy alone, but rare-class estimates remain high-variance.
Validation rigor	ViT/Swin use StratifiedGroupKFold within development data and keep a fixed lesion-disjoint test untouched until final evaluation.	This is the strongest model-selection protocol among the image-only notebooks.
GNN value proposition	Adds structured spatial nodes, broad ablation, bootstrap uncertainty and node-level SHAP/LIME, but final Macro F1 is below ViT/Swin.	Its main contribution is pipeline analysis/interpretability rather than superior predictive performance in this run.
AUC reporting caveat	Two CNN notebooks have failed aggregate AUC calculation; ViT main summary has NaN despite valid class-wise ROC values.	Do not copy aggregate AUC fields blindly. The table distinguishes unavailable, directly reported and derived AUC values.

Bottom line

For a clean paper/report comparison, use two layers of evidence: (1) a strict same-test comparison of ViT vs Swin vs GNN on the 1,506-image seed-171 test, and (2) a separate same-split comparison of EfficientNet-B2 vs ResNet50 on the 1,494-image seed-42 test. Do not present one global five-model rank without stating this protocol difference.

Scope: This PDF reflects the code, markdown, stored textual/numeric outputs and reported analysis in the five uploaded notebooks. No external benchmark values were inserted. Missing or failed notebook outputs are labeled rather than imputed.